
Algorithm Particle Swarm Optimization

Set the algorithm's parameters and input the data
Generate the initial population
Generate the initial velocity values randomly
Record the best solution observed thus far G_{best}
while the termination criterion is not met **do**
 for each particle in the population **do**
 Record the best position observed by the particle $P_{best,particle}$
 end for
 if the new best position observed in the process **then**
 Update G_{best}
 end if
 for each particle in the population **do**
 Compute the particle's velocity
 Update the particle's position
 end for
 Update inertia weight parameter
end while
Report G_{best} as the solution
